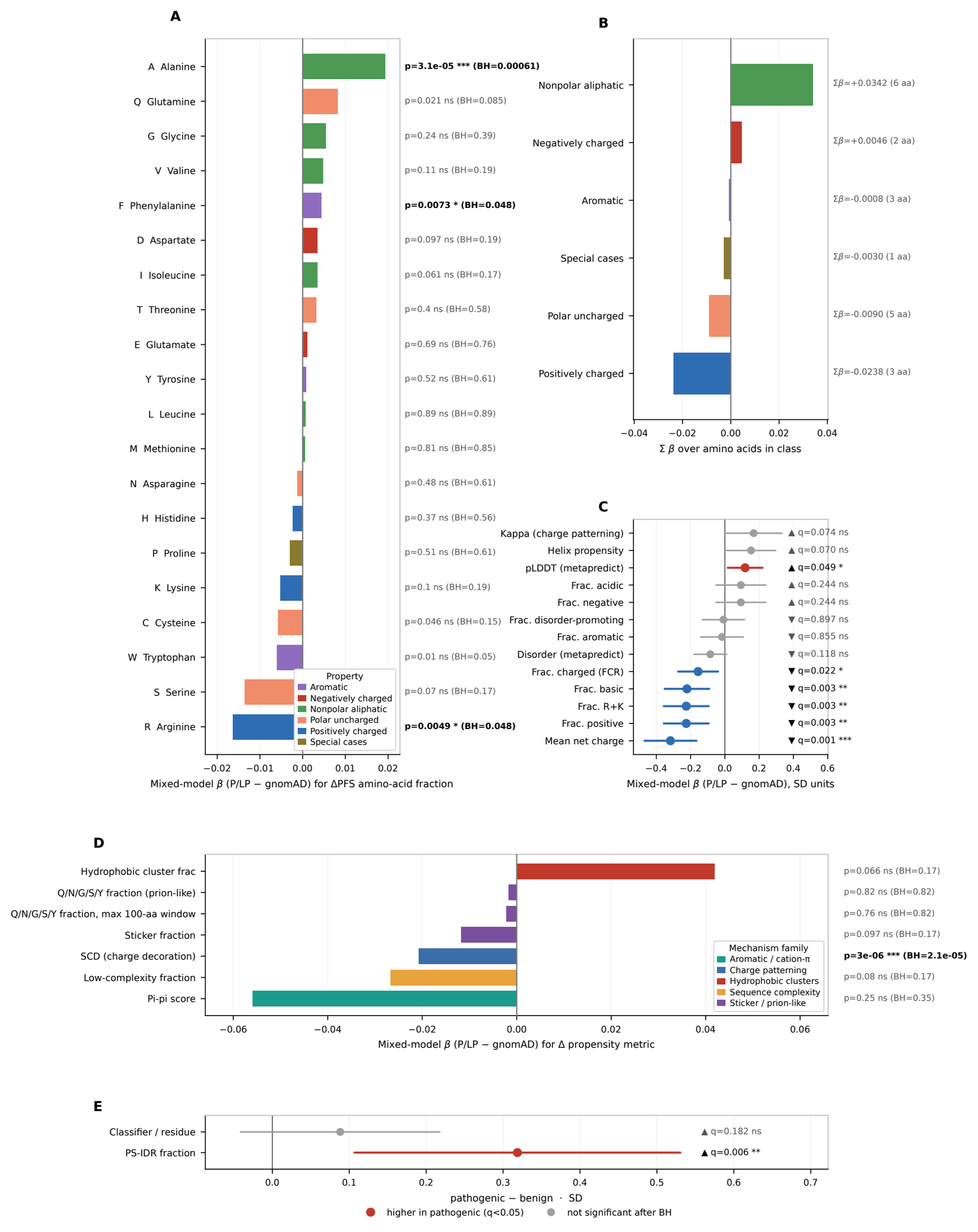
A
A Alanine
Q Glutamine
G Glycine
V Valine
F Phenylalanine
D Aspartate
I Isoleucine
T Threonine
E Glutamate
Y Tyrosine
L Leucine
M Methionine
N Asparagine
H Histidine
P Proline
K Lysine
C Cysteine
W Tryptophan
S Serine
R Arginine
p=3.1e-05 *** (BH=0.00061)
p=0.021 ns (BH=0.085)
p=0.24 ns (BH=0.39)
p=0.11 ns (BH=0.19)
p=0.0073 * (BH=0.048)
p=0.097 ns (BH=0.19)
p=0.061 ns (BH=0.17)
p=0.4 ns (BH=0.58)
p=0.69 ns (BH=0.76)
p=0.52 ns (BH=0.61)
p=0.89 ns (BH=0.89)
p=0.81 ns (BH=0.85)
p=0.48 ns (BH=0.61)
p=0.37 ns (BH=0.56)
p=0.51 ns (BH=0.61)
p=0.1 ns (BH=0.19)
p=0.046 ns (BH=0.15)
p=0.01 ns (BH=0.05)
p=0.07 ns (BH=0.17)
p=0.0049 * (BH=0.048)
Property
Aromatic
Negatively charged
Nonpolar aliphatic
Polar uncharged
Positively charged
Special cases
−0.02 −0.01 0.00 0.01 0.02
Mixed-model β (P/LP − gnomAD) for ΔPFS amino-acid fraction
B
Nonpolar aliphatic
Negatively charged
Aromatic
Special cases
Polar uncharged
Positively charged
Σβ=+0.0342 (6 aa)
Σβ=+0.0046 (2 aa)
Σβ=-0.0008 (3 aa)
Σβ=-0.0030 (1 aa)
Σβ=-0.0090 (5 aa)
Σβ=-0.0238 (3 aa)
−0.04 −0.02 0.00 0.02 0.04
Σ β over amino acids in class
C
Kappa (charge patterning)
Helix propensity
pLDDT (metapredict)
Frac. acidic
Frac. negative
Frac. disorder-promoting
Frac. aromatic
Disorder (metapredict)
Frac. charged (FCR)
Frac. basic
Frac. R+K
Frac. positive
Mean net charge
▲ q=0.074 ns
▲ q=0.070 ns
▲ q=0.049 *
▲ q=0.244 ns
▲ q=0.244 ns
▼ q=0.897 ns
▼ q=0.855 ns
▼ q=0.118 ns
▼ q=0.022 *
▼ q=0.003 **
▼ q=0.003 **
▼ q=0.003 **
▼ q=0.001 ***
−0.4 −0.2 0.0 0.2 0.4 0.6
Mixed-model β (P/LP − gnomAD), SD units
D
Hydrophobic cluster frac
Q/N/G/S/Y fraction (prion-like)
Q/N/G/S/Y fraction, max 100-aa window
Sticker fraction
SCD (charge decoration)
Low-complexity fraction
Pi-pi score
p=0.066 ns (BH=0.17)
p=0.82 ns (BH=0.82)
p=0.76 ns (BH=0.82)
p=0.097 ns (BH=0.17)
p=3e-06 *** (BH=2.1e-05)
p=0.08 ns (BH=0.17)
p=0.25 ns (BH=0.35)
Mechanism family
Aromatic / cation-π
Charge patterning
Hydrophobic clusters
Sequence complexity
Sticker / prion-like
−0.06 −0.04 −0.02 0.00 0.02 0.04 0.06
Mixed-model β (P/LP − gnomAD) for Δ propensity metric
E
Classifier / residue
PS-IDR fraction
▲ q=0.182 ns
▲ q=0.006 **
0.0 0.1 0.2 0.3 0.4 0.5 0.6 0.7
pathogenic − benign · SD
higher in pathogenic (q<0.05)
not significant after BH